

SURAJ PATEL

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Contact : [Gmail](#) / [LinkedIn](#) / [Github](#) / [Leetcode](#)

EDUCATION

Bachelor of Technology, Acropolis Institute of Technology & Research 2023 – 2027
Branch: Computer Science and Engineering (AI&ML)

Secondary School, Jayant Public School, Malikhedi 2022 – 2023

High School, St. Thomas H.S. School, Kshipra 2020 – 2021

PROJECTS

LegalQuery - Legal RAG Chatbot (BNS) - [Project link](#)

Technologies: Python, LangChain, Qdrant, Llama 3.3 (Groq API), HuggingFace BGE Embeddings, Streamlit, FastAPI, RAGAS

- Built a RAG-based legal chatbot for Indian criminal law using vector retrieval and multi-source search.
- Implemented document ingestion with recursive chunking and FlashrankRerank for improved retrieval quality.
- Added Qdrant semantic caching (cosine similarity 0.7) to reduce redundant LLM calls and improve latency.
- Evaluated performance using RAGAS metrics and deployed a Streamlit-based real-time legal assistant.

Financial News Sentiment Analysis - [Project link](#)

Technologies: Python, NumPy, Pandas, Scikit-Learn, PyTorch, HuggingFace Transformers, Streamlit, Joblib

- Built and benchmarked TF-IDF + Logistic Regression, Bi-LSTM, and fine-tuned BERT models on 4,845 financial news headlines.
- Addressed class imbalance using stratified splits, class weighting, and Macro-F1 evaluation; implemented negation-aware preprocessing and n-gram TF-IDF.
- Achieved 74% test accuracy with baseline models and deployed production-ready inference using Streamlit.

Video Summarization System - [Project link](#)

Technologies: Python, openai-whisper, Qwen (GGUF), llama-cpp-python, HuggingFace Hub, PyTorch, Streamlit

- Developed a pipeline that converts YouTube URLs or uploaded videos into structured summaries using Whisper ASR and Qwen GGUF models.
- Designed adaptive summarization strategies: Direct (< 10 min), Hierarchical (10–30 min), and Map-Reduce (> 30 min) to prevent context overflow and reduce token load.

SKILLS

Technical Skills	Machine Learning, Deep Learning, Natural Language Processing, Data Structure and Algorithm, Object-Oriented Programming, Competitive Programming
Languages	C/C++, Java, Python
Frameworks	NumPy, Pandas, Matplotlib, Seaborn, Tensorflow, LangChain, LangGraph, FastAPI
Tools	GitHub, VS Code, Colab, Jupyter Notebook, Kaggle, Docker

TECHNICAL ACHIEVEMENTS

- Solved 250+ Data Structures and Algorithms (DSA) problems on LeetCode and GeeksforGeeks

EXTRACURRICULAR

- * Conducted a 4-week robotics workshop **RoboLearn** for school students, introducing basic robotics concepts.
- * Organized and coordinated an **IEEE Congress Meet** as an IEEE Member.